NAMAL UNIVERSITY, MIANWALI

Department of Computer Science

CSC-271 – Database Systems

Complex Computing Problem – Milestone 2

Restaurant Management Information System (RMIS)

Entity-Relationship Design Report

Table 1: Project Submission Details

Field	Details
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Course	CSC-271 – Database Systems
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Table 2: Group Members Information

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1. Introduction

The Restaurant Management Information System (RMIS) is a comprehensive, multi- branch database-driven solution designed to digitize and streamline all core operational aspects of a modern restaurant business. The system addresses the growing complexity of managing food-service enterprises that operate across multiple physical locations, each with its own staff, inventory, pricing structure, and order-processing workflow.

In the contemporary food-service industry, restaurants face significant challenges in coordinating branch-level operations, maintaining consistent service quality, and ensuring accurate data flows between the kitchen, the front-of-house, human resources, and management. Manual or fragmented systems often result in order discrepancies, inventory mismanagement, payroll errors, and a lack of visibility across branches.

RMIS resolves these challenges by providing a unified relational database backend that tracks every transactional and operational event within the business. The system handles customer-facing operations such as order placement and discount application, as well as back-end functions including staff scheduling, salary disbursement, inventory restocking, and menu pricing management.

This report constitutes Milestone 2 of the Complex Computing Problem (CCP) for CSC-271 – Database Systems. It presents a thorough analysis of the entities in the system, their attributes, the business rules that govern their behavior, and the relationships between them. The culmination of this analysis is a well-structured Entity-Relationship Diagram (ERD) that serves as the logical blueprint for the system's database design.

2. Problem Statement

Multi-branch restaurant operations generate vast amounts of interrelated data daily – from order transactions and menu pricing to employee shift management and inventory consumption. Without a centralized, well-structured database system, restaurants face the following operational problems:

- Inability to accurately track orders across different branches, leading to billing errors and customer dissatisfaction.
- There is no systematic link between menu items and their ingredient requirements, leading to inventory shortages and food waste.
- Difficulty in managing employee roles, shift schedules, and salary records in a transparent and auditable manner.
- Lack of time-bound pricing control, making it difficult to apply promotional or

seasonal price changes without data corruption.

- Absence of a structured discount management mechanism, resulting in inconsistent application of offers across branches.
- No reliable mechanism for tracking which manager oversees which staff members, compromising accountability.

The RMIS database system addresses all of the above problems by defining clear entities, relationships, and business constraints that ensure data integrity, operational consistency, and management transparency across all branches.

3. Objectives

3.1 General Objective

To design a normalized, well-structured relational database schema for a multi-branch Restaurant Management Information System that supports all core operational domains of the business.

3.2 Specific Objectives

1. To identify and define all primary entities that constitute the restaurant system, including Branch, Users, Orders, MenuItem, Inventory, Shift, Salary, Discount, Price, OrderItem, Category, and MenuIngredient.
2. To enumerate all relevant attributes for each entity, distinguishing primary keys, foreign keys, and descriptive attributes.
3. To establish and document all inter-entity relationships with precise cardinalities (one-to-one, one-to-many, or many-to-many) and minimum/maximum participation constraints.
4. To define clear business rules that govern data integrity and operational logic across all system domains.
5. To construct a complete and professionally formatted Entity-Relationship Diagram (ERD) that accurately represents the system architecture.
6. To produce a report that conforms to academic standards suitable for submission as part of the CSC-271 Complex Computing Problem milestones.

4. System Overview

The RMIS is modeled around the concept of a restaurant chain with multiple branches. Each branch operates independently but shares a common schema for data management. The system encompasses four main operational domains:

- **Operations Domain:** Covers the lifecycle of customer orders – from placement to payment – including discount application and order item detail tracking.
- **Menu & Inventory Domain:** Manages the restaurant's food offerings (MenuItems organized by Category), their ingredient compositions (MenuIngredient), and the raw material stock (Inventory) held at each branch.
- **Human Resources Domain:** Tracks all staff members (Users) including their roles, branch assignments, managerial hierarchy, shift scheduling (Shift), and salary records (Salary).
- **Pricing Domain:** Maintains a time-bound pricing history for each menu item through the Price entity, enabling effective promotional and seasonal pricing management.

The system supports role-based access and operations, where a user (employee) may act as a cashier (takes orders), kitchen staff (prepares orders), or manager (manages staff, shifts, and salaries). This dual-role capability is modeled through the multi-relationship structure of the Users entity.

5. Entity Identification & Attribute Definition

The following entities were identified from the ERD. Each entity represents a distinct, real-world object within the RMIS domain. Entities are presented with their complete attribute sets, data types, and key designations.

Key: **[PK]** = Primary Key **[FK]** = Foreign Key

5.1 Branch

Represents a physical restaurant location. Each branch is uniquely identified and stores its contact and operational details.

Branch

Attribute	Data Type	Description / Key
BranchId [PK]	INT	Primary Key – unique identifier for each branch
BranchName	VARCHAR(100)	Name of the branch location
Address	TEXT	Full physical address of the branch
PhoneNumber	VARCHAR(20)	Contact phone number
Email	VARCHAR(100)	Official email address
OpeningTime	TIME	Daily opening time of the branch
ClosingTime	TIME	Daily closing time of the branch
IsActive	BOOLEAN	Indicates whether the branch is currently operational

5.2 Users

Represents all staff members employed at the restaurant. Users can hold multiple roles (cashier, kitchen staff, manager) and are linked to a specific branch.

Users		
Attribute	**Data Type**	**Description / Key**
Username [PK]	VARCHAR(50)	Primary Key – unique login identifier
Name	VARCHAR(100)	Full name (First + Last) of the employee
Email	VARCHAR(100)	Employee email address
PhoneNumber	VARCHAR(20)	Employee contact number
Password	VARCHAR(255)	Encrypted password for authentication
Role	VARCHAR(30)	Job role (e.g., Manager, Cashier, Chef)
IsActive	BOOLEAN	Active employment status
HireDate	DATE	Date the employee was hired
ResignationDate	DATE	Date of resignation (nullable)
ManagedBy [FK]	VARCHAR(50)	FK referencing Username of the managing user

CreatedAt	DATETIME	Record creation timestamp

5.3 Orders

Records each customer order transaction. An order is taken by one user (cashier) and prepared by another (kitchen staff), and may optionally have a discount applied.

Orders		
Attribute	**Data Type**	**Description / Key**
OrderId [PK]	INT	Primary Key – unique order identifier
TakenBy [FK]	VARCHAR(50)	FK – Username of the cashier who took the order
PreparedBy [FK]	VARCHAR(50)	FK – Username of the kitchen staff who prepared it
DateTime	DATETIME	Timestamp of the order
Status	VARCHAR(20)	Current status (e.g., Pending, Completed, Cancelled)
PaymentStatus	VARCHAR(20)	Payment state (e.g., Paid, Unpaid)
CashTendered	DECIMAL(10,2)	Amount of cash received from customer
ChangeDue	DECIMAL(10,2)	Change to return to the customer
TotalAmount	DECIMAL(10,2)	Total monetary value of the order

5.4 OrderItem

A weak/associative entity that represents each line item within an order. It links a specific menu item to an order and captures quantity and unit price at the time of the transaction.

OrderItem		
Attribute	**Data Type**	**Description / Key**
OrderItemId [PK]	INT	Primary Key – unique order item identifier

OrderId [FK]	INT	FK – References the parent order
MenuItemId [FK]	INT	FK – References the ordered menu item
QuantityOrdered	INT	Number of units ordered for this item
UnitPriceAtOrder	DECIMAL(10,2)	Price per unit captured at order time
Subtotal	DECIMAL(10,2)	Calculated: QuantityOrdered × Unit-PriceAtOrder

5.5 MenuItem

Represents a food or beverage product offered by the restaurant. Each menu item belongs to a category and can have multiple price records and ingredient compositions.

MenuItem		
Attribute	**Data Type**	**Description / Key**
MenuItemId [PK]	INT	Primary Key – unique menu item identi-fier
MenuItemType	VARCHAR(30)	Type classification (e.g., Food, Beverage, Dessert)
ItemName	VARCHAR(100)	Display name of the menu item
Description	TEXT	Textual description of the item
CategoryId [FK]	INT	FK – Category this item belongs to
IsAvailable	BOOLEAN	Whether the item is currently offered
CreatedAt	DATETIME	Record creation timestamp

5.6 Category

Groups menu items into logical classifications such as Starters, Main Course, Beverages, or Desserts.

Category

Attribute	Data Type	Description / Key
CategoryId [PK]	INT	Primary Key – unique category identifier
CategoryName	VARCHAR(50)	Name of the category (e.g., Main Course)

5.7 Price

Maintains a historical and current pricing record for each menu item. Multiple price records may exist per item to accommodate promotional or seasonal pricing. Only one record is active at any time.

Price		
Attribute	**Data Type**	**Description / Key**
PriceId [PK]	INT	Primary Key – unique price record identifier
MenuItemId [FK]	INT	FK – Menu item this price applies to
Amount	DECIMAL(10,2)	Price value in currency units
EffectiveDate	DATE	Date from which this price is valid
EndDate	DATE	Date when this price expires (nullable for active)
IsActive	BOOLEAN	Flags the currently active price
CreatedAt	DATETIME	Record creation timestamp

5.8 Inventory

Tracks raw material stock at each branch. Each inventory item has a reorder level to trigger restocking and a unit cost for purchase records.

Inventory		
Attribute	**Data Type**	**Description / Key**

InventoryId [PK]	INT	Primary Key – unique inventory item identifier
InventoryName	VARCHAR(100)	Name of the raw material or ingredient
Quantity	DECIMAL(10,2)	Current stock quantity on hand
Unit	VARCHAR(20)	Unit of measurement (e.g., kg, litre, piece)
ReorderLevel	DECIMAL(10,2)	Minimum quantity before restocking is triggered
UnitCost	DECIMAL(10,2)	Cost per unit of the inventory item
LastRestocked	DATETIME	Timestamp of the most recent restock
BranchId [FK]	INT	FK – Branch where this inventory is held

5.9 MenuIngredient

An associative entity linking menu items to their required inventory ingredients. It records the quantity of each ingredient needed to prepare one unit of a menu item.

MenuIngredient		
Attribute	**Data Type**	**Description / Key**
MenuIngredientId [PK]	INT	Primary Key – unique mapping identifier
MenuItemId [FK]	INT	FK – Menu item that requires this ingredient
InventoryId [FK]	INT	FK – Inventory item (ingredient) required
QuantityRequired	DECIMAL(10,2)	Amount of the ingredient needed per unit of menu item

5.10 Shift

Records individual work shifts assigned to employees. A shift captures the date, timing, type, and status of a scheduled or completed work period.

Shift		
Attribute	**Data Type**	**Description / Key**
ShiftId [PK]	INT	Primary Key – unique shift identifier
UserId [FK]	VARCHAR(50)	FK – Username of the employee assigned to this shift
ShiftDate	DATE	Calendar date of the shift
StartTime	TIME	Scheduled start time
EndTime	TIME	Scheduled or actual end time
ShiftType	VARCHAR(20)	Type of shift (e.g., Morning, Evening, Night)
Status	VARCHAR(20)	Shift status (e.g., Scheduled, Completed, Absent)
CreatedBy [FK]	VARCHAR(50)	FK – Username of the manager who created the shift
CreatedAt	DATETIME	Timestamp when the shift was created

5.11 Salary

Records salary disbursement events for each employee. Each record corresponds to a single payment transaction, managed by an authorized manager.

Salary		
Attribute	**Data Type**	**Description / Key**
SalaryId [PK]	INT	Primary Key – unique salary record identifier
UserId [FK]	VARCHAR(50)	FK – Username of the employee receiving the salary
Amount	DECIMAL(10,2)	Salary amount disbursed in this payment
PaymentDate	DATE	Date the salary was paid

Notes	TEXT	Optional remarks (e.g., bonus, deduction details)

5.12 Discount

Defines time-bound discount policies that can be applied to orders. Discounts have a defined active period, controlled by start and end dates.

Discount		
Attribute	**Data Type**	**Description / Key**
DiscountId [PK]	INT	Primary Key – unique discount identifier
DiscountName	VARCHAR(100)	Name/label of the discount offer
DiscountValue	DECIMAL(5,2)	Discount percentage or flat amount
StartDate	DATE	Date from which the discount becomes active
EndDate	DATE	Date after which the discount expires
IsActive	BOOLEAN	Flag indicating current activation status

6. Business Rules

The following business rules define the operational constraints and logical governance of the RMIS database system. These rules are directly derivable from the ERD and are critical to maintaining data integrity.

6.1 Branch & User Rules

- A Branch must have at least one User assigned to it; a User must belong to exactly one Branch.
- A User can be managed by at most one other User (supervisor), forming a self-referencing hierarchy within the Users entity.
- A User may have one of several roles: Manager, Cashier, or Kitchen Staff. Role determines which operations the user is authorized to perform.
- When a User is deactivated (IsActive = FALSE), their shifts and salary records

are retained for historical auditing.

6.2 Order & Order Item Rules

- Every Order must be taken by exactly one User (cashier) and prepared by at most one User (kitchen staff). An order cannot exist without a taker.
- An Order may optionally be associated with a Discount. At most one discount can be applied per order.
- Every Order must contain at least one OrderItem. An OrderItem cannot exist without a parent Order.
- The UnitPriceAtOrder in OrderItem is captured at the time of the transaction and is independent of any subsequent price changes to the MenuItem.
- The TotalAmount in Orders must equal the sum of all OrderItem Subtotals minus any applicable discount.

6.3 Menu & Pricing Rules

- Every MenuItem must belong to exactly one Category. A Category may contain zero or more MenuItems.
- A MenuItem may have multiple Price records over time, but only one Price record with IsActive = TRUE is permitted at any given time.
- A Price record with a NULL EndDate is considered currently active. When a new price is activated, the previous active record's EndDate must be set.
- A MenuItem may require zero or more Inventory ingredients (via MenuIngredient). The same Inventory item can be a required ingredient for multiple MenuItems.

6.4 Inventory Rules

- Each Inventory item is held at exactly one Branch. A Branch may hold many inventory items.
- When the Quantity of an Inventory item falls at or below the ReorderLevel, the system should flag it for restocking.
- The Inventory entity tracks raw materials; it does not track finished menu items.

6.5 Shift & Salary Rules

- Every Shift must be assigned to exactly one User. A User may have many Shifts over time.
- A Shift must be created by exactly one User (manager). The CreatedBy field references the managing user.
- Every Salary record must be associated with exactly one User. A User may receive multiple Salary payments over time.
- Salary management is performed by authorized Users (managers) as indicated by the Manages relationship.

7. Relationships and Cardinality

The table below summarizes all relationships identified in the ERD, including relationship names, participating entities, cardinality type, and a brief description.

Entity A	Relationship	Entity B	Cardinality	Description
Branch	**Has**	Users	1 : M (Mand. : Opt.)	One branch employs one or more users; each user works at exactly one branch.
Branch	**Has_StockTe**	Inventory	1 : M (Mand. : Mand.)	One branch holds one or more inventory items; each inventory item belongs to exactly one branch.
Users	**Takes**	Orders	1 : M (Mand. : Opt.)	One user (cashier) can take many orders; each order is taken by exactly one user.
Users	**Prepares**	Orders	1 : M (Opt. : Opt.)	One user (kitchen staff) can prepare many orders; each order may be prepared by one user.

Entity A	Relationship	Entity B	Cardinality	Description
Users	**Manages**	Salary	1 : M (Mand. : Opt.)	One user (manager) manages salary records; each salary record is administered by one manager.
Users	**Receives**	Salary	1 : M (Mand. : Mand.)	One user receives one or more salary payments; each salary record belongs to one user.
Users	**Creates**	Shift	1 : M (Mand. : Opt.)	One manager creates many shifts; each shift is created by exactly one user.
Users	**Works**	Shift	1 : M (Mand. : Mand.)	One user is assigned to many shifts; each shift belongs to exactly one user.
Users	**ManagedBy**	Users	1 : M (Opt. : Opt.)	Self-referencing: one user (manager) supervises many users; each user has at most one manager.
Orders	**Has**	Discount	M : 1 (Opt. : Opt.)	Many orders can use one discount; a discount may be applied to many orders or none.
Orders	**Contain**	OrderItem	1 : M (Mand. : Mand.)	One order contains one or more order items; each order item belongs to exactly one order.
OrderItem	**References**	MenuItem	M : 1 (Mand. : Mand.)	Many order items can reference one menu item; a menu item can appear in many order items.
MenuItem	**Contains**	Category	M : 1 (Mand. : Mand.)	Many menu items belong to one category; a category can hold many menu items.

Entity A	Relationship	Entity B	Cardinality	Description
MenuItem	**Has_Price**	Price	1 : M (Mand. : Mand.)	One menu item has one or more price records; each price record belongs to one menu item.
MenuItem	**Requires**	MenuIngredi	1 : M (Opt. : Mand.)	One menu item requires one or more ingredients; each MenuIngredient record belongs to one item.
MenuIngred	**ContainsTex**	Inventory	M : 1 (Mand. : Opt.)	Many MenuIngredient records reference one inventory item; an inventory item may be used in many recipes.

8. ERD Description & Data Flow

The Entity-Relationship Diagram for the RMIS captures all entities, their attributes, and inter-entity relationships using standard ERD crow's foot notation. The diagram is organized into four logical zones:

8.1 Core Operational Zone (Centre)

The Orders entity is positioned at the center of the diagram, reflecting its role as the primary transactional entity. It connects outward to Users (via *Takes* and *Prepares* relationships), to Discount (via *Has*), and downward to OrderItem (via *Contain*). OrderItem further references MenuItem, which in turn connects to Category, Price, and MenuIngredient.

8.2 Human Resources Zone (Right Side)

The Branch entity anchors the upper-right region. Below it, Users represents all employees, with self-referential management links (*ManagedBy*). Two sub-structures branch from Users: the Shift entity (for scheduling) and the Salary entity (for payroll). The *Manages* relationship connects Users to Salary, while *Works* and *Creates* connect Users to Shift.

8.3 Inventory & Supply Zone (Left Side)

The Inventory entity on the left side is linked to Branch via the *Has StockText* rela- tionship, indicating branch-level stock. It is also referenced by MenuIngredient, which bridges inventory to the menu system. This bi-directional link enables the system to calculate ingredient consumption per order.

8.4 Pricing & Menu Zone (Bottom)

The MenuItem entity serves as the hub of the lower section, connecting to Category (via *Contains*), to Price (via *Has_Price*), and to MenuIngredient (via *Requires*). This structure enables multi-dimensional querying: finding all items in a category, retrieving current and historical prices, and determining ingredient requirements for any menu item.

9. Conclusion

This report has presented a thorough and structured analysis of the Restaurant Management Information System as represented by the Entity-Relationship Diagram submitted for Milestone 2 of CSC-271.

Twelve distinct entities have been identified and documented: Branch, Users, Orders, OrderItem, MenuItem, Category, Price, Inventory, MenuIngredient, Shift, Salary, and Discount. Each entity has been described with its complete attribute set, with clear designation of primary keys and foreign keys. The system's sixteen relationships have been cataloged with precise cardinality specifications and business-rule justifications.

The ERD demonstrates a well-normalized design that avoids data redundancy while preserving all necessary semantic relationships between business entities. Notably, the design supports advanced operational features such as historical price tracking (via the Price entity), recipe-level ingredient management (via MenuIngredient), and organizational hierarchy modeling (via the self-referencing ManagedBy relationship in Users).

The logical design presented in this milestone forms a solid foundation for the subsequent milestone, wherein this ERD will be translated into a physical relational schema with appropriate SQL DDL statements, normalization verification, and constraint definitions.

A. AI Tool Disclosure

In compliance with the course instructions, the following AI tools and prompts were used in the preparation of this report:

Tool Used

- Claude (Anthropic) – claude.ai

Prompts Used

1. *"You are an expert technical report writer and database analyst. I will provide you with a milestone instructions PDF and an ERD image. Generate a complete, professional, and well-structured report based strictly on the format file and the ERD. Analyze all entities, attributes, relationships and cardinalities from the ERD."*
2. *"Identify all entities from the ERD image and list their attributes including data types and whether they are primary or foreign keys."*
3. *"List all relationships from the ERD with their cardinality types (one-to-one, one- to-many, many-to-many), minimum and maximum participation constraints, and relationship names."*
4. *"Write business rules for the Restaurant Management Information System based on the ERD provided."*

All AI-generated content was reviewed, verified against the ERD, and edited for accuracy and academic appropriateness by the team members before inclusion in this report.

— End of Report —